Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER PROJECT REVIEW

Date: 06-05-2026

Design and develop a React JS based web application for Ticket Booking of an Internal Department Event

SDG: SDG 9 Industry, Innovation and Infrastructure

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Courses Handling Faculty:

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INTRODUCTION

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1. Traditional campus event management relies on notice boards, manual registrations, and paper tracking, which are inefficient and error-prone.
2. These outdated methods often cause poor communication, data loss, and difficulty in monitoring participation.
3. In today's digital era, automation is essential to streamline event handling and improve efficiency.
4. The Smart Campus Event Management System addresses these challenges by providing a centralized digital platform.
5. It allows students to explore upcoming events, register online, and share feedback seamlessly.
6. Administrators can manage event details, track registrations, and analyze participation statistics effectively.
7. The system is built with Spring Boot, Spring MVC, and Spring Data JPA for backend robustness.
8. Thymeleaf, HTML5, and CSS3 power the interactive frontend, ensuring scalability and user-friendliness.

PROBLEM STATEMENT

In many colleges and universities, the management of campus events such as workshops, seminars, and cultural activities is still handled using manual or semi-digital methods. Information about events is often shared through notice boards, emails, or messaging platforms, which can lead to miscommunication, missed updates, and limited student engagement. Additionally, event registration is frequently done through paper forms or unstructured online tools, making it difficult to maintain accurate records and track participation.

For administrators, managing multiple events becomes challenging due to the lack of a centralized system. Tasks such as creating events, updating schedules, handling registrations, and analyzing participation data are time-consuming and prone to errors. There is also limited capability to filter events or generate meaningful insights like registration statistics.

OBJECTIVE(S)

- **To design and develop** a Smart Campus Event Management System that enables efficient creation, management, and organization of campus events through a user-friendly web interface for both students and administrators.
- **To implement** a secure and automated event registration and tracking system that allows students to register easily and helps administrators monitor participation accurately and transparently.
- **To improve** the overall event management process by reducing manual effort, minimizing errors, and enhancing accessibility through a centralized digital platform for managing events, registrations, and feedback.

PROPOSED SYSTEM

1. The proposed system is a web-based Smart Campus Event Management System that provides a centralized platform for managing all campus events efficiently.
2. It allows students to browse, register, and view upcoming events through an intuitive and user-friendly interface.
3. The system enables administrators to create, update, and delete events using secure login and role-based access control.
4. It includes automated registration management, ensuring accurate tracking of participants and eliminating manual errors.
5. The system provides advanced search and filtering options based on date, department, and event type for easy navigation.
6. It generates event participation statistics and feedback reports, helping administrators analyze and improve future events.

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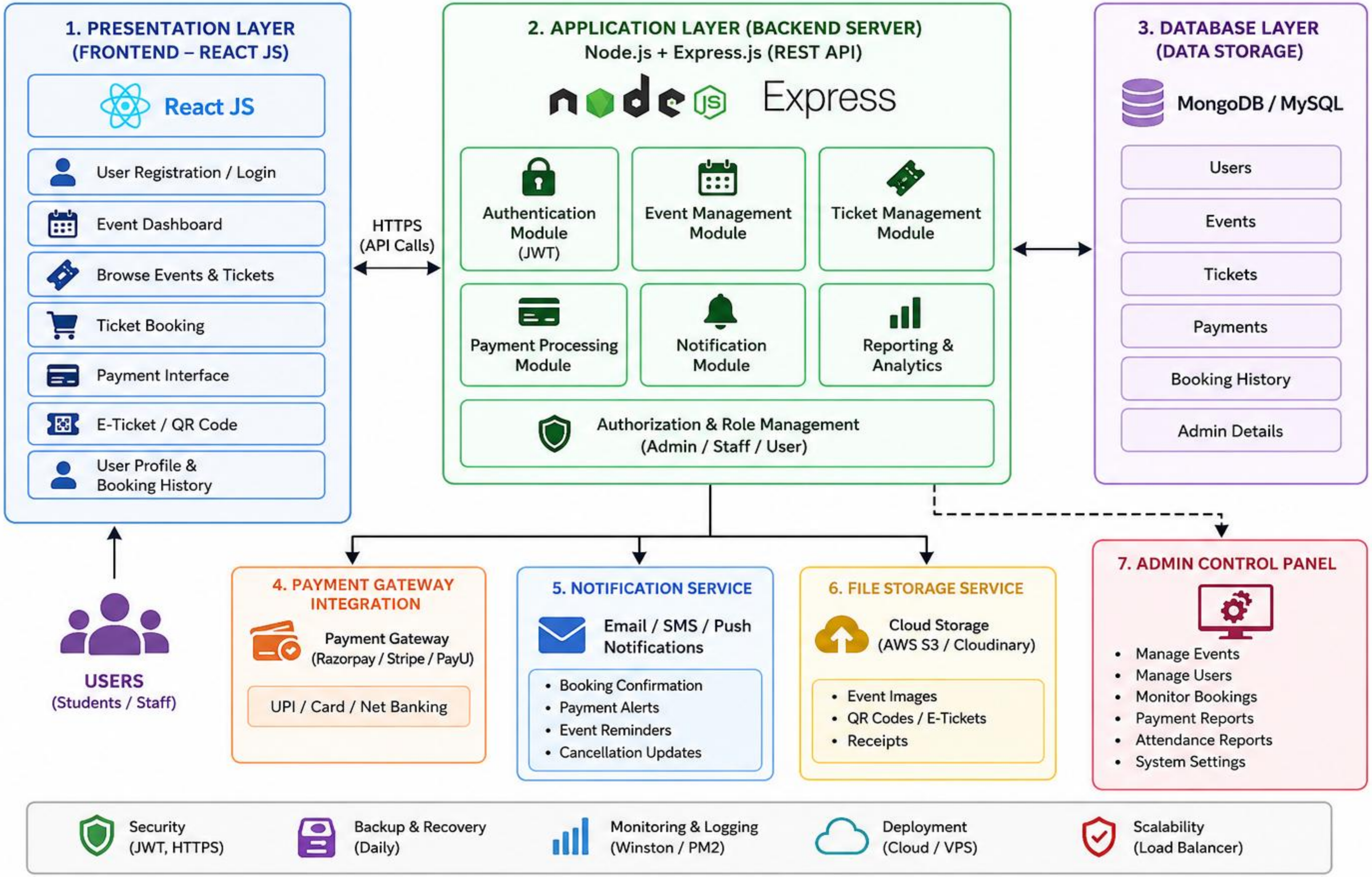
PROPOSED SYSTEM ARCHITECTURE

Ticket Booking System for Internal Department Event



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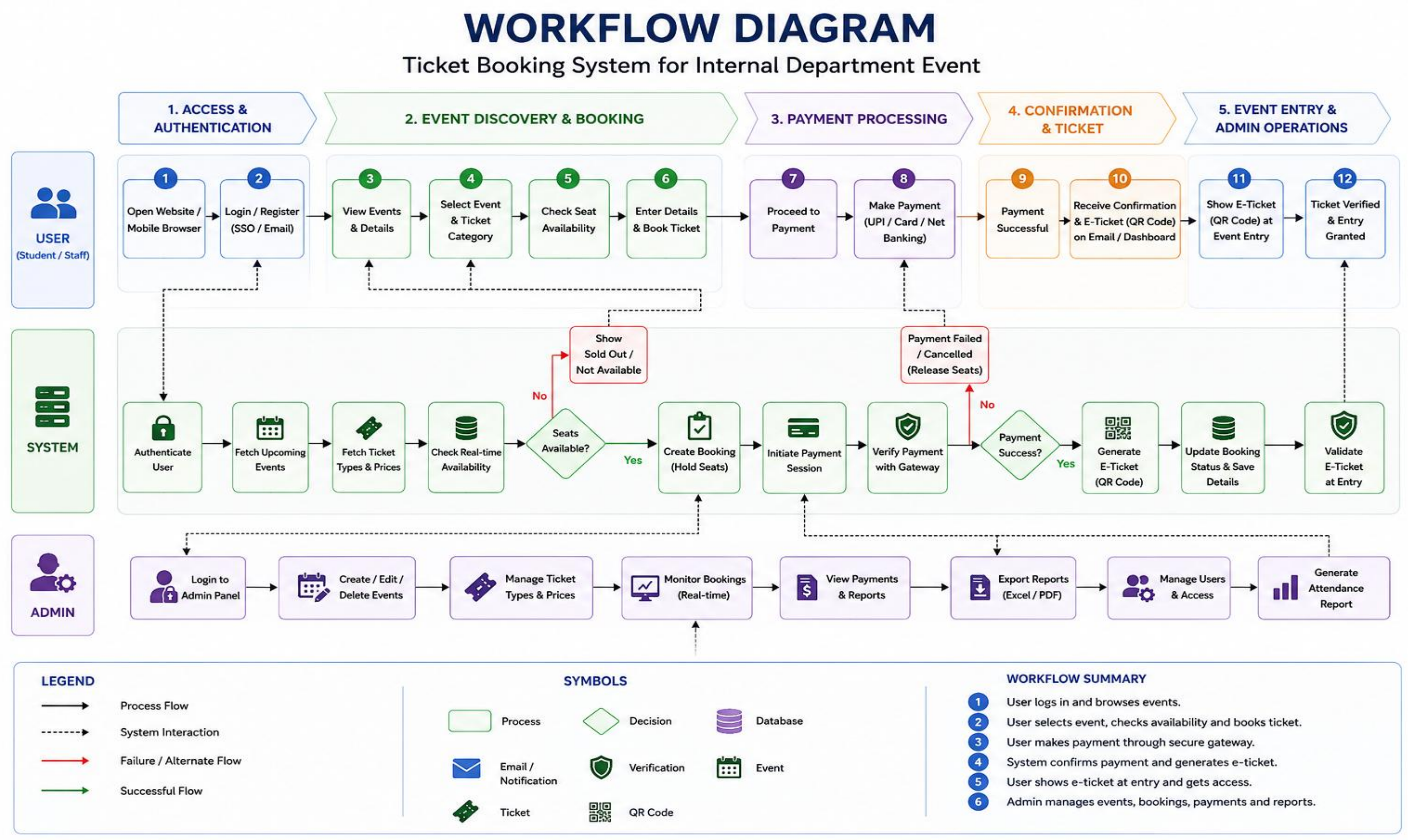

WORKFLOW DIAGRAM
Ticket Booking System for Internal Department Event
1. ACCESS & AUTHENTICATION
2. EVENT DISCOVERY & BOOKING
3. PAYMENT PROCESSING
4. CONFIRMATION & TICKET
5. EVENT ENTRY & ADMIN OPERATIONS
USER (Student / Staff)
1 Open Website / Mobile Browser
2 Login / Register (SSO / Email)
3 View Events & Details
4 Select Event & Ticket Category
5 Check Seat Availability
6 Enter Details & Book Ticket
7 Proceed to Payment
8 Make Payment (UPI / Card / Net Banking)
9 Payment Successful
10 Receive Confirmation & E-Ticket (QR Code) on Email / Dashboard
11 Show E-Ticket (QR Code) at Event Entry
12 Ticket Verified & Entry Granted
SYSTEM
Authenticate User
Fetch Upcoming Events
Fetch Ticket Types & Prices
Check Real-time Availability
Seats Available?
No
Show Sold Out / Not Available
Yes
Create Booking (Hold Seats)
Initiate Payment Session
Verify Payment with Gateway
Payment Success?
No
Payment Failed / Cancelled (Release Seats)
Yes
Generate E-Ticket (QR Code)
Update Booking Status & Save Details
Validate E-Ticket at Entry
ADMIN
Login to Admin Panel
Create / Edit / Delete Events
Manage Ticket Types & Prices
Monitor Bookings (Real-time)
View Payments & Reports
Export Reports (Excel / PDF)
Manage Users & Access
Generate Attendance Report
LEGEND
Process Flow
System Interaction
Failure / Alternate Flow
Successful Flow
SYMBOLS
Process
Decision
Database
Email / Notification
Verification
Event
Ticket
QR Code
WORKFLOW SUMMARY
1 User logs in and browses events.
2 User selects event, checks availability and books ticket.
3 User makes payment through secure gateway.
4 System confirms payment and generates e-ticket.
5 User shows e-ticket at entry and gets access.
6 Admin manages events, bookings, payments and reports.

MODULE DESCRIPTION

Module 1: User Management
Handles student and admin registration and login functionalities with secure authentication. It ensures **role-based access control**, where students can access event-related features and administrators can manage the system. This module maintains user data privacy and system security.

Module 2: Event Management (Core Processing)
Implements the core functionality of the system, including **event creation, updating, and deletion** by admins. It also allows students to **browse events, register for events, and submit feedback**. The module ensures smooth handling of event workflows and user interactions.

Module 3: Database Management
Responsible for storing and managing all system data such as **event details, student information, registrations, and feedback**. It ensures data consistency, integrity, and efficient retrieval using a structured database system like MySQL or H2.

Module 4: Reporting & Analytics
Generates reports on **event participation, registration counts, and feedback analysis**. It provides useful insights to administrators, helping them evaluate event success and make better decisions for future events.

TECHNOLOGIES USED

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Frontend:

- React 19 (Vite)
- Plain CSS (App.css / index.css) with animations and gradients
- Google Fonts — Poppins
- No UI library — fully custom components

Backend:

- Node.js with Express.js
- REST API (port 5000)
- Nodemon (dev auto-restart)

Database:

- SQLite via sql.js (file-based:
- bookings.db
-)
- 3 tables: students, events, bookings

TECHNOLOGIES USED

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Third-Party APIs:

Nodemailer + Gmail SMTP — OTP email & booking confirmation emai
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Tools:

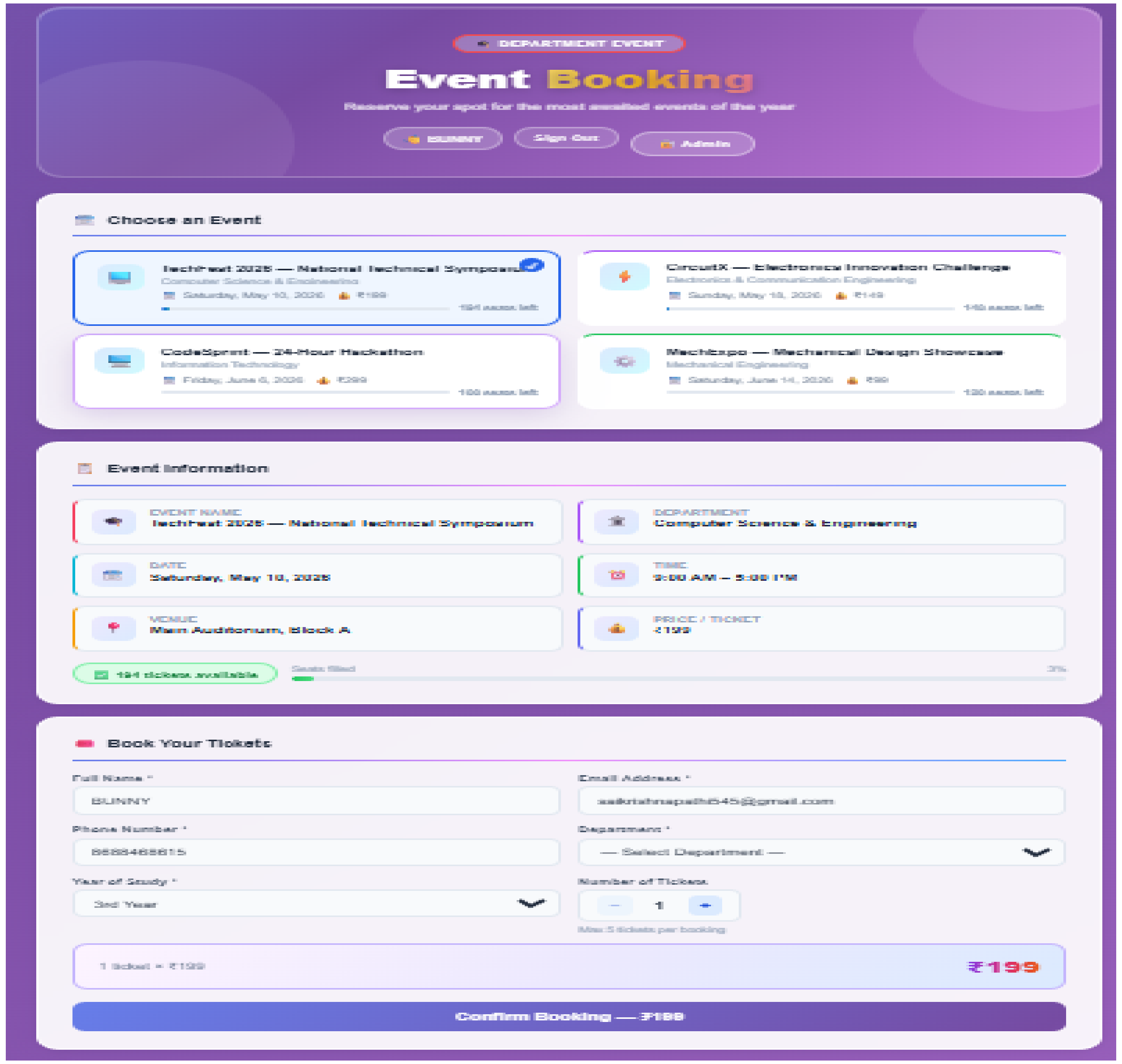
- Vite — frontend build & dev server
- dotenv — environment variable management
- ESLint — code linting
- Git — version control

Hardware / Software Requirements:

- OS: Windows (tested), macOS/Linux compatible
- Node.js v18 or higher
- npm v9 or higher
- Modern browser (Chrome, Edge, Firefox)
- Internet connection (for Gmail SMTP and Fast2SMS)
- Gmail account with App Password enabled (2FA required)
- Minimum 4GB RAM, any modern CPU

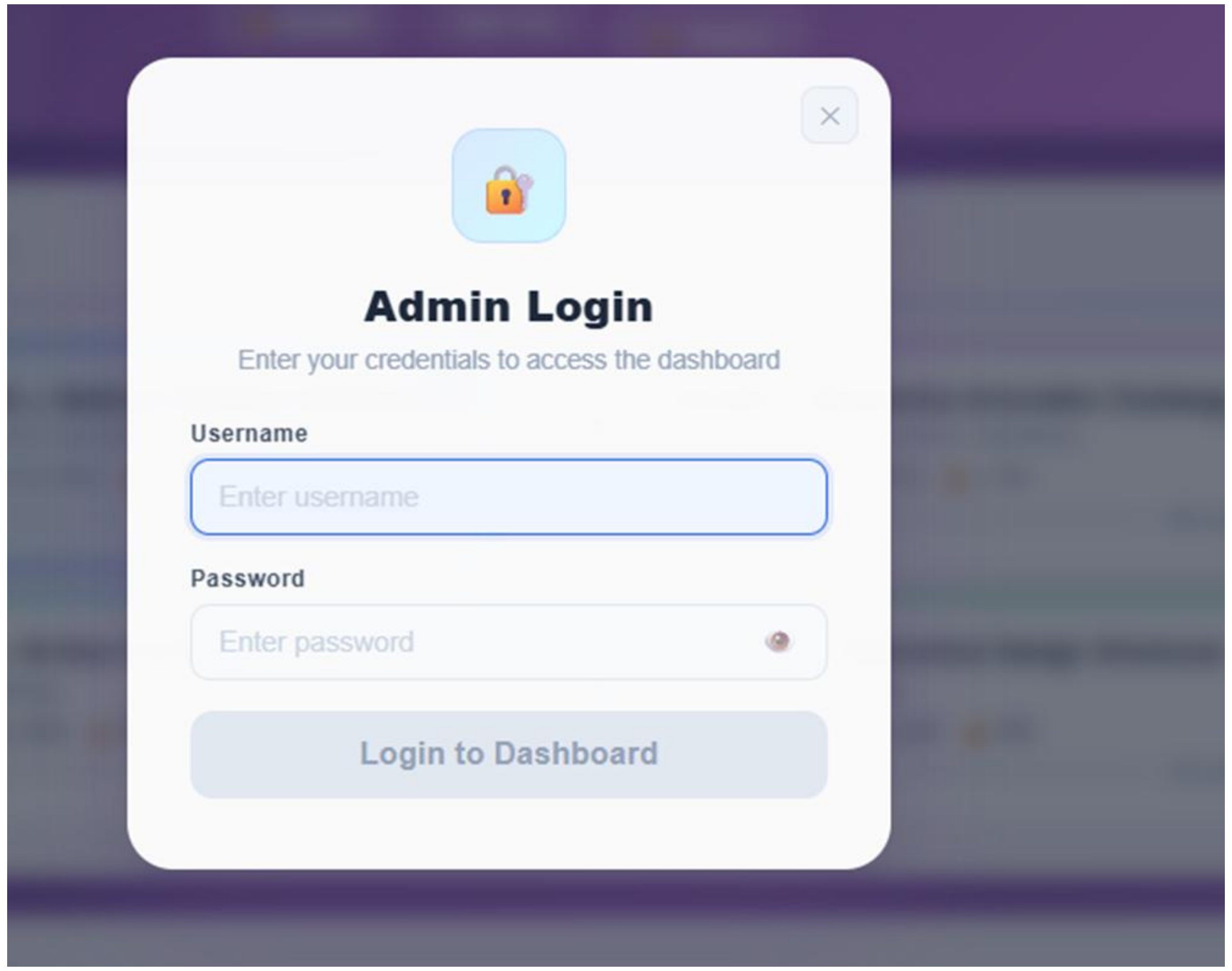
IMPLEMENTATION SCREENSHOTS

Dashboard:

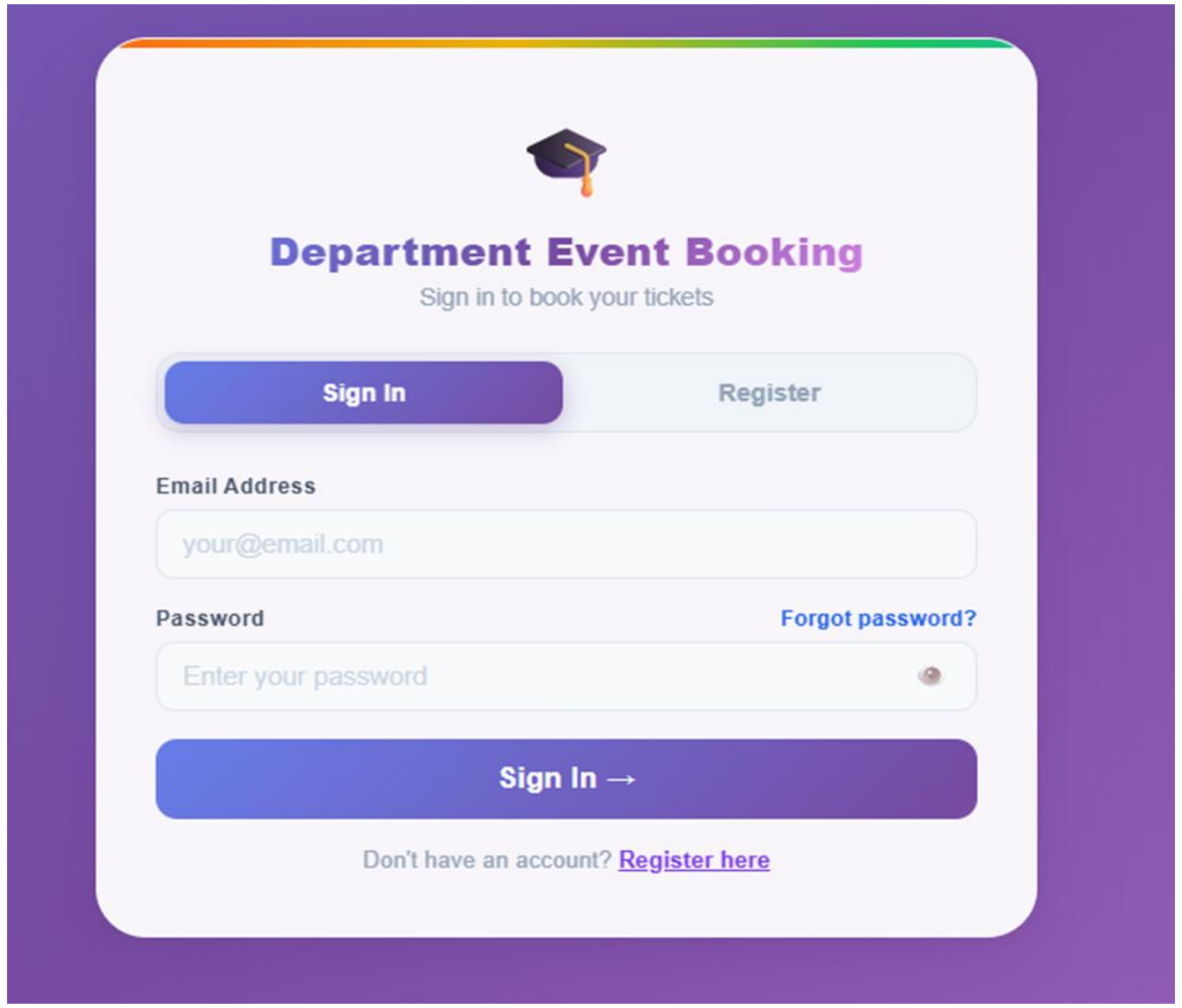


IMPLEMENTATION SCREENSHOTS

- Admin Login Page



- User Login Page



IMPLEMENTATION SCREENSHOTS

Student Table:

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: students

	id	full_name	email	phone	department	year	password	created_at
1	1	sai	sai@edu.in	8688468615	Computer Science (CSE)	3rd Year	1234567	17 Apr 2026, 9:54 am
2	2	sai	sai@email.com	8688468615	Civil Engineering (CIVIL)	4th Year	12345678	17 Apr 2026, 10:45 am
3	3	sathwik	sathwik@gmail.com	9154525601	Computer Science (CSE)	3rd Year	12345678	17 Apr 2026, 10:52 am
4	4	sai	sai@gmail.com	9154525601	Mechanical Engineering (MECH)	3rd Year	123456w	4 May 2026, 11:41 pm
5	5	BUNNY	saikrishnapathi545@gmail.com	8688468615	Computer Science (CSE)	3rd Year	1234567	4 May 2026, 11:55 pm
6	6	rajii	raji@gmail.com	9492859848	Mechanical Engineering (MECH)	4th Year	567890	5 May 2026, 3:57 pm

Events table:

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: events

	id	name	department	date	time	venue
1	1	TechFest 2026 — National Technical …	Computer Science & Engineering	Saturday, May 10, 2026	9:00 AM – 5:00 PM	Main Auditorium, Bloc
2	2	CircuitX — Electronics Innovation …	Electronics & Communication …	Sunday, May 18, 2026	10:00 AM – 4:00 PM	ECE Seminar Hall, Blo
3	3	CodeSprint — 24-Hour Hackathon	Information Technology	Friday, June 6, 2026	8:00 AM – 8:00 AM (Next Day)	IT Lab Complex, Block
4	4	MechExpo — Mechanical Design Showcase	Mechanical Engineering	Saturday, June 14, 2026	9:30 AM – 3:30 PM	Workshop Hall, Block

IMPLEMENTATION SCREENSHOTS

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Bookings Table:

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: bookings

	id	booking_id	event_id	student_id	name	email	phone	department	year	ticket_count	total_amount	
1	1	TF-PSFSMEH4	1	NULL	sai	sai@edu.in	8688468615	Computer Science (CSE)	3rd Year	1	199	17 A
2	2	TF-5CKH1FP8	1	NULL	Sai krishna	sai@edu.in	8688468615	Computer Science (CSE)	3rd Year	2	398	17 A
3	3	TF-VFZARH7S	1	NULL	sathwik	sathwik@gmail.com	9154525601	Computer Science (CSE)	3rd Year	2	398	17 A
4	4	TF-H6HB741M	1	NULL	sai	sai@gmail.com	9154525601	Mechanical Engineering (MECH)	3rd Year	1	199	4 Ma
5	5	TF-WVXU00LV	2	NULL	BUNNY	saikrishnapathi5...	8688468615	Computer Science (CSE)	3rd Year	2	298	6 Ma

New Database | Open Database | Write Changes | Re

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: sqlite_sequence

	name	seq
1	events	4
2	students	6
3	bookings	5

ADVANTAGES & LIMITATIONS

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Advantages:

Centralized Management – All event-related activities (creation, registration, feedback) are handled in one platform, reducing confusion and duplication.

Time-Saving & Efficient – Automates event registration and management, minimizing manual work and speeding up processes.

User-Friendly Interface – Students can easily browse and register for events, improving participation and engagement.

Accurate Data Handling – Reduces human errors through automated validation and database management.

Insightful Analytics – Provides reports and statistics on event participation, helping administrators make better decisions.

Limitations :

Internet Dependency – Requires stable internet access for both students and administrators.

Initial Setup Complexity – Development and configuration of the system may require technical expertise.

Security Risks – If not properly secured, the system may be vulnerable to unauthorized access or data breaches.

Limited Offline Access – Users cannot access features without being connected to the system online.

Maintenance Required – Needs regular updates and monitoring to ensure smooth performance and security.

Thank you